

Day 13: Cloud Database Migration & Scalable Backend Hardening

Key Milestone: Neon Serverless PostgreSQL & Async Backend Deployment

Sprint Focus: Execution of core deliverables for Day 13 across AI/NLP, Header Forensics, IOC Extraction, Graph Correlation, Case Management, and UI Integration.

Daily Deliverables: Neon Cloud PostgreSQL Instance, Async SQLAlchemy Connection Layer, Database Performance Benchmarks.

Individual Team Member Task Breakdown & Contributions

Team Member & Lead Role	Domain & Ownership Area	Tasks Executed & Deliverables on Day 13
Muniswami.Y AI / NLP Lead	Explainable AI, Intent & Phishing Detection, Social Engineering signals	Hardened NLP model execution against adversarial payload injection and extremely long email header string overflows.
Jeevan reddy.G Header & Forensics Lead	Email header analysis, SPF/DKIM/DMARC, Mail path reconstruction, Timestamp analysis	Configured SQLAlchemy async engine with `asyncpg` connection pooling and SSL mode sanitization for cloud PostgreSQL.
Lahari.K IOC & Infrastructure Lead	URL, Domain, IP extraction, Geolocation, ASN/ISP mapping, Malicious indicator detection	Added database indexes on `origin_ip`, `case_id`, `sender_domain`, and `created_at` for high-speed cross-case searching.
Hemanth.P Graph & Campaign Intelligence Lead	Attack Relationship Graph, Similarity detection, Campaign grouping & correlation	Optimized graph serialization algorithms to stream large nodal structures in sub-50ms JSON responses.
Abbas.N Risk, Case & Response Lead	Dynamic Threat Score engine, Automatic Forensic Case creation, Recommended Response actions	Provisioned serverless Neon.tech PostgreSQL database (`threattrace_db`) and executed production schema migrations.
Moulaanbee.N Frontend + Assistant + Simulator Lead	Dashboard, Visualizations (Graph, Map, Timeline), AI Chatbot, Email Threat Simulator, Overall UI/UX	Updated API client services to dynamically fallback between live cloud endpoints and local development mocks.

Integration, Verification & Quality Assessment

Cross-Module Integration:	Successfully migrated all local SQLite records to live Neon Cloud PostgreSQL; verified zero connection pooling dropouts.
Quality Assurance Status:	All code unit tests and interface contracts for Day 13 verified clean. Git commit pushed with zero blocking errors.